Certified low-T-count multipliers for binary fields from Chinese remaindering

Claude Opus 5.5 Extra Effort[1]

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We report ancilla-free Clifford+T circuits for multiplication in GF(2^n) for thirteen field sizes, $n = 5$–8, 10–16, 32 and 64, released as independently checkable gate lists. Where a published decomposition with correct released data exists, ours use fewer T gates: 57 against 59 for $n = 5$, 69 against 77 for $n = 6$, 113 against 123 for $n = 8$ and 151 against 157 for $n = 10$. For $n = 7$ we use 97 against the reported 101. For $n \geq 11$ the only reported values, 215 to 3729 T gates, come from decompositions that, we show, do not decompose GF(2^n) multiplication; our circuits use 179 to 1255. The circuits are built from a composition principle. If the product in GF(2^n) can be computed by Chinese remaindering in a product of small algebras A_i, then the T-count is at most $\sum_i \mathcal{T}(A_i)$, because every block is a linear relabelling of a small multiplier. The blocks can be subfields, local rings $\mathbb{F}_2[y]/(q^e)$ or the points of an algebraic curve. With optimized blocks, for example 15 T gates for $\mathbb{F}_2[y]/(y^2)$ (optimal for circuits of this form, by exhaustive search) and 35 for GF(4)$[\epsilon]/(\epsilon^2)$, the principle gives certified circuits for prime n, where subfield interpolation does not apply. A genus-2 curve with 33 rational points over GF(16) gives $n = 64$. FastTODD-type reduction and simulated annealing over the coset of decompositions remove further gates. For $n = 5$, 6 and 7, annealing with reheating and slow cooling takes the composed starts from 72 to 58 (then 57 by reduction), from 85 to 69 and from 111 to 97 T gates. A check of the released data behind the published values shows that one family of recent records, which holds the reported values for $n = 5$ and $n \geq 11$, decomposes a bilinear map with zero divisors rather than GF(2^n) multiplication. Exhaustive search shows that 17 T gates are optimal for GF(4) among circuits of this form. We could not reproduce the published 23 T gates for GF(8), which does not refute that value. Every circuit was certified by an exact verifier, by independent implementations, and by simulation of all inputs for each record with $n \leq 12$. We release the thirteen decompositions together with explicit gate lists, their Clifford corrections, and a standalone checker. The results were found by a multi-agent language-model system working against a frozen verifier.

I. INTRODUCTION

In fault-tolerant quantum computing with magic-state distillation, the T gate is the expensive gate, and its count is a standard cost metric [1, 2]. Minimizing the T-count of a circuit is hard in general. For circuits whose non-Clifford part is a phase polynomial it is equivalent to decoding a punctured Reed–Muller code, or to finding a minimal symmetric (Waring) decomposition of a third-order tensor over $\mathbb{F}_2$ [3, 4]. Heuristics for this problem include TODD [4] and its descendants FastTODD [5], reinforcement learning (AlphaTensor-Quantum) [6], tensor-decomposition methods [7], and LLM-guided evolutionary search (VarTODD+GigaEvo) [8]. Together they define the state of the art on standard benchmark circuits [9].

Multiplication in GF(2^n) is one of these benchmarks and a building block of quantum attacks on binary elliptic curves [10–12]. Its best reported T-counts (Table I) come from three sources. Vandaele reports 23 for $n = 3$, from FastTODD applied to a Karatsuba-type circuit with ancillas [5, 13]; we could not reproduce this value (Sec. VI). AlphaTensor-Quantum gives the best values for $n = 4$, 6 and 8 [6], which Fisher *et al.* match. VarTODD+GigaEvo reports the best values for $n = 5$, 7, 9–16, 32 and 64 [8]. Its values for $n = 7$–10 start from decompositions of Ref. [7]; the others start from matrices derived from Vandaele's circuits. We checked the released data behind these values (Sec. VI). The Khoruzhii-start results are correct GF(2^n) multipliers. The Vandaele-start ones, including those for $n = 5$ and $n \geq 11$, are not: their tensors have zero divisors.

These methods treat each GF(2^n) as an unstructured tensor. We use the algebra instead. Classical algebraic complexity computes a product of polynomials by reducing it modulo several coprime polynomials, multiplying the residues and recombining them (Toom–Cook, Karatsuba, Winograd) [14–16]. Chudnovsky and Chudnovsky extended the idea to the points of algebraic curves [18]. We show that the T-count composes in the same way (Theorem 1). Every residue product is a linear relabelling of a multiplier in a small algebra, so the T-count of a composed circuit is the sum of the T-counts of its blocks. Optimized blocks then give good large circuits without any search on the large tensor, and FastTODD-type reduction or annealing improves them further.

An earlier version of this work used only subfield interpolation and covered six composite n. Here, residues modulo powers of irreducible polynomials (local rings) extend the method to prime n, an algebraic curve extends it to $n = 64$, and annealing improves the small fields $n = 6$ and 7. Section II fixes notation. Section III proves the composition principle, and Section IV gives its three instances. Section V describes the reduction and annealing. Section VI presents the records and the small-field status. Section VII describes the verification, and Section VIII how the results were found.

TABLE I. T-counts of ancilla-free GF(2^n) multipliers. "Reported" is the best published value: Ref. [8] (Table I) for $n \neq 6, 8$, and Refs. [6, 8] for $n = 6$ and 8. †The released decomposition and its start are not a GF(2^n) multiplication tensor (Sec. VI). ‡Not released; the released start of the same pipeline has the same defect. "Checked" is the best published value whose released decomposition we verified: Ref. [6] for n = 5–8, and Ref. [8] (Khoruzhii start) for $n = 10$. For $n = 7$, Ref. [7] also releases a 101-gate decomposition, which we did not check. "Start" is the composed circuit of Sec. IV: K is the base field and the blocks are listed; their T-counts are in Table II, and "Raw" is their sum. Here ∞^e, t^e are rational places of multiplicity e, $p = y^2 + y + 1$, "deg k" is an irreducible place of degree k (a block GF(2^{mk})), and ω generates GF(4). "Search" is the step from the start to the final count: FastTODD-type reduction (FT) or annealing (SA) (Sec. V).

n	Reported	Checked	This work	K	Start: blocks	Raw	Search
5	49†	59	**57**	$\mathbb{F}_2$	$\infty^3, x^2, (x+1)^2$, deg 2	72	SA 60, 58; FT 57
6	77	77	**69**	GF(4)	5 points: 5×17	85	FT 79, SA 77, 71, 69
7	101	104	**97**	$\mathbb{F}_2$	$\infty^3, x^3, (x+1)^2$, deg 2, deg 3	111	SA 100, 99, 97
8	123	123	**113**	GF(16)	3 points (Karatsuba): 3×39	117	FT
10	157	157	**151**	GF(4)	$\infty^2, 0^2, 1^2, \omega^2, \bar{\omega}$	157	FT
11	215†	–	**179**	$\mathbb{F}_2$	$\infty^3, x^2, (x+1)^2, p^2$, 2×deg 3, deg 4	187	FT 181, 179
12	243†	–	**191**	GF(16)	5 points: 5×39	195	FT
13	293†	–	**219**	$\mathbb{F}_2$	$\infty^3, x^2, (x+1)^2, p^2$, 2×deg 3, 2×deg 4	226	FT
14	319†	–	**229**	GF(4)	∞, $0^2, 1^2, \omega^2, \bar{\omega}^2$, 2×deg 2	235	FT
15	363†	–	**253**	GF(8)	9 points: 9×29	261	FT
16	383†	–	**267**	GF(16)	7 points: 7×39	273	FT
32	1177†	–	**571**	GF(16)	15 points: 15×39	585	FT
64	3729‡	–	**1255**	GF(16)	genus-2 curve, 33 points: 33×39	1287	FT

II. PRELIMINARIES

Let $P \in \mathbb{F}_2[x]$ be irreducible of degree n. Elements $a, b, c \in \mathrm{GF}(2^n) = \mathbb{F}_2[x]/(P)$ are coordinate vectors in $\mathbb{F}_2^n$. The in-place multiplier $|a, b, c\rangle \mapsto |a, b, c \oplus ab\rangle$ equals $H_c\, D\, H_c$, where H_c applies Hadamard gates to the c register and D is the diagonal gate

$$D = \sum_x (-1)^{f_n(x)} |x\rangle\langle x|, \qquad f_n(a, b, c) = c \cdot (ab), \quad (1)$$

where $x = (a, b, c)$ and $\cdot$ is the dot product over $\mathbb{F}_2$. The function f_n is a cubic form in the $N = 3n$ variables of x, whose monomials are $a_i b_j c_k$ for each x^k that appears in $x^{i+j} \bmod P$.

A T gate applied to the parity $u \cdot x$ (a phase gadget, conjugated by CNOT gates) contributes the phase $\omega^{\ell_u(x)}$, with $\omega = e^{i\pi/4}$ and $\ell_u(x) \in \{0, 1\}$ the parity as an integer. A set $\{u_1, \ldots, u_m\} \subset \mathbb{F}_2^N \setminus \{0\}$ *implements* D if D equals $\prod_r \omega^{\ell_{u_r}}$ times a diagonal Clifford gate. The T-count $\mathcal{T}(n)$ is the least such m. It does not depend on P or on the basis, because those changes are CNOT circuits.

Throughout, T-counts refer to this model: ancilla-free circuits $H_c D H_c$ in which D is a phase polynomial built from CNOT, T and diagonal Clifford gates. Every circuit reported here has this form, and every optimality statement below holds within it. Such statements are not lower bounds for circuits with intermediate Hadamard gates, measurements or classical feedforward. We also count only T gates. A lower T-count does not by itself imply a lower runtime or physical-qubit cost, and we have not optimized the CNOT count (Appendix A).

Lemma 1 (Signature-tensor criterion [3, 4])**.** *$\{u_r\}$ implements D if and only if every variable and every pair of variables lies in the support of an even number of the u_r, and every triple lies in an odd number of them exactly when it is a monomial of f_n. Equivalently, $\sum_r u_r^{\otimes 3} \equiv S_n \pmod 2$, where S_n is the zero-diagonal symmetric tensor of f_n.*

Proof. Over $\mathbb{Z}_8$, $\ell_u(x) = \sum_{i \in u} x_i - 2\sum_{i<j \in u} x_i x_j + 4\sum_{i<j<k \in u} x_i x_j x_k$, and terms of degree ≥ 4 vanish because their coefficients are multiples of 8. Hence $4f_n - \sum_r \ell_{u_r}$ has linear coefficients $-\#\{r \ni i\}$, quadratic coefficients $2\#\{r \ni i, j\}$ and cubic coefficients $4(f_{ijk} - \#\{r \ni i, j, k\})$. A diagonal phase $\omega^{g(x)}$ is Clifford exactly when $g = 2h$ with h of degree ≤ 2 over $\mathbb{Z}_4$ and even quadratic coefficients (products of S and CZ gates). That is, the singles and pairs must be even and the triples must match $f_n \bmod 2$. □

The criterion is exact, so a set of vectors that passes it is a proof that the circuit is correct for all inputs; no sampling is involved. The same definitions apply to any finite commutative $\mathbb{F}_2$-algebra A with a chosen basis: $f_A(x, y, z) = z \cdot (xy)$ is a cubic form in $3 \dim A$ variables, and $\mathcal{T}(A)$ is its T-count. A change of basis, or of the nondegenerate pairing used for z, is an invertible linear substitution, so $\mathcal{T}(A)$ depends only on the isomorphism class of A. For example, $\mathcal{T}(\mathrm{GF}(2^k)) = \mathcal{T}(k)$.

III. COMPOSITION PRINCIPLE

Lemma 2 (Pull-back)**.** *Let $\{u_r\}$ implement the cubic form $g(y)$ on $\mathbb{F}_2^M$, and let $L : \mathbb{F}_2^N \to \mathbb{F}_2^M$ be linear. Then $\{L^{\mathsf{T}} u_r\}$, after deleting zero vectors and cancelling equal pairs, implements $g(Lx)$. Its size is at most the size of $\{u_r\}$.*

Proof. $\ell_u(Lx) = \ell_{L^{\mathsf{T}}u}(x)$, and Clifford gates on $y = Lx$ pull back to Clifford gates in x. Zero vectors and pairs u, u contribute Clifford phases ($\omega^{2\ell_u} = i^{\ell_u}$). □

Theorem 1. *Let $A_1, \dots, A_s$ be finite commutative $\mathbb{F}_2$-algebras. Suppose there are $\mathbb{F}_2$-linear maps $\alpha_i, \beta_i : \mathrm{GF}(2^n) \to A_i$ and $\Gamma : A_1 \times \cdots \times A_s \to \mathrm{GF}(2^n)$ such that*

$$ab = \Gamma\big(\alpha_1(a)\beta_1(b), \dots, \alpha_s(a)\beta_s(b)\big) \quad \text{for all } a, b. \tag{2}$$

Then $\mathcal{T}(n) \le \sum_i \mathcal{T}(A_i)$.

Proof. Write $\Gamma(p_1, \dots, p_s) = \sum_i \Gamma_i(p_i)$. For each i the map $p \mapsto c \cdot \Gamma_i(p)$ is $\mathbb{F}_2$-linear in $p \in A_i$, so it equals $\gamma_i(c) \cdot p$ for an $\mathbb{F}_2$-linear $\gamma_i : \mathbb{F}_2^n \to A_i$. Hence $f_n(a, b, c) = \sum_i f_{A_i}\big(\alpha_i(a), \beta_i(b), \gamma_i(c)\big)$. Lemma 2, applied to each term with $L_i = (\alpha_i, \beta_i, \gamma_i)$, gives a decomposition of size $\le \sum_i \mathcal{T}(A_i)$. The union of the pulled-back sets implements the sum, because phases multiply and Clifford gates compose. □

The bilinear complexity of multiplication over $\mathbb{F}_2$ is linear in n [18, 19], and $\mathcal{T} \le 7\times$(number of CCZ gates), so $\mathcal{T}(n) = O(n)$ was known. The point of Theorem 1 is the constant: it sums the blocks' *T-counts*, not $7\times$ their bilinear ranks. For GF(16) the bilinear rank is 9, so $7 \times 9 = 63$ against $\mathcal{T}(4) = 39$.

IV. THREE FAMILIES OF BLOCKS

Places of the projective line. Let $K = \mathrm{GF}(2^m)$ and $n = md$, and write $\mathrm{GF}(2^n) = K[y]/(g)$ with g irreducible of degree d over K. Elements are polynomials A, B of degree $< d$, and $C = AB$ has degree $\le 2d - 2$. Choose pairwise coprime powers $q_i^{e_i}$ of monic irreducible $q_i \in K[y]$ and a multiplicity $e_\infty \ge 0$ at infinity, with

$$e_\infty + \sum_i e_i \deg q_i \;\ge\; 2d - 1. \tag{3}$$

The residue $C \bmod q_i^{e_i}$ is the product of $A \bmod q_i^{e_i}$ and $B \bmod q_i^{e_i}$ in the ring $K[y]/(q_i^{e_i})$. The top e_∞ coefficients of C depend only on the top e_∞ coefficients of A and B, and they are their product in $K[y]/(y^{e_\infty})$ after reversing the coefficient order. These residues determine C: if they all vanish, C is divisible by $\prod_i q_i^{e_i}$ and has degree $\le 2d - 2 - e_\infty < \sum_i e_i \deg q_i$, so $C = 0$. Recovering C and reducing it modulo g is K-linear, which gives Eq. (2) with the blocks $K[y]/(q_i^{e_i})$ and $K[y]/(y^{e_\infty})$. Hence

$$\mathcal{T}(md) \le \mathcal{T}\big(K[y]/(y^{e_\infty})\big) + \sum_i \mathcal{T}\big(K[y]/(q_i^{e_i})\big). \tag{4}$$

TABLE II. Block algebras and their T-counts $\mathcal{T}(A)$. "Exact" values come from exhaustive enumeration of the coset of decompositions (2^{22} elements). The others are certified upper bounds. $\mathcal{T}(\mathrm{GF}(8)) = 29$ is the best value we could certify; the published 23 is discussed in Sec. VI.

A	$\dim_{\mathbb{F}_2} A$	$\mathcal{T}(A)$	source
$\mathbb{F}_2$	1	7	CCZ, exact
$\mathbb{F}_2[y]/(y^2)$	2	15	this work, exact
$\mathbb{F}_2[y]/(y^3)$	3	25	this work
GF(4)	2	17	Refs. [5, 6]; exact (this work)
GF(8)	3	29	Ref. [6]
GF(16)	4	39	Refs. [6, 8]
$\mathrm{GF}(4)[\epsilon]/(\epsilon^2)$	4	35	this work
GF(64)	6	69	this work
$\mathrm{GF}(2^7)$	7	97	this work
GF(256)	8	113	this work

A place of multiplicity one is a field, $K[y]/(q) \cong \mathrm{GF}(2^{m \deg q})$. A rational place of multiplicity two over K is the ring of dual numbers $K[\epsilon]/(\epsilon^2)$, and $\mathbb{F}_2[y]/(p^2)$ with $p = y^2 + y + 1$ is isomorphic to $\mathrm{GF}(4)[\epsilon]/(\epsilon^2)$.

With rational places of multiplicity one only, Eq. (4) is Toom–Cook interpolation: if $2^m \ge 2d - 2$, the $2d - 2$ finite points and infinity give $\mathcal{T}(md) \le (2d - 1)\mathcal{T}(m)$, and $d = 2$ is Karatsuba's formula for every m [15–17]. Interpolation gives the starts for $n = 6$, 8, 12, 15, 16 and 32 in Table I. Prime n has no proper subfield, so $K = \mathbb{F}_2$, which has only three rational places (0, 1, ∞). For $K = \mathbb{F}_2$, the local rings make the difference. Each unit of degree costs 7.5 T gates in $\mathbb{F}_2[y]/(y^2)$ (15 for degree 2) but 8.5 in the field GF(4) (17 for degree 2). For the same reason, the dual numbers over GF(4) (35 T gates for 2 units) are cheaper than a degree-2 place over GF(4), the field GF(16) (39 for 2 units). A small dynamic program over the place multiset, with the block costs of Table II, chooses the starts for $n = 5$, 7, 10, 11, 13 and 14.

An algebraic curve for $n = 64$. Over $K = \mathrm{GF}(16)$ with $d = 16$, Eq. (3) needs 31 units, but the projective line has only 17 rational places. The remaining 14 units must come from seven degree-2 places, each a GF(256) block of 113 T gates, so $17 \times 39 + 7 \times 113 = 1454$. Chudnovsky and Chudnovsky replace the line by a curve with more points [18]. The genus-2 curve $C : \; y^2 + y = x^5$ over GF(16) has 33 rational points. For each $x \in \mathrm{GF}(16)$, x^5 lies in GF(4), whose elements have trace zero in $\mathrm{GF}(16)/\mathbb{F}_2$, so each x gives two points; the point at infinity is the 33rd. This attains the Hasse–Weil bound $q + 1 + 2g\sqrt{q} = 33$. Let $D = 16P_\infty + P_a$ with P_a an affine rational point. By Riemann–Roch, $\dim L(D) = 16$ and $\dim L(2D) = 33 = 2d + g - 1$. For a suitable place Q of degree 16, whose residue field is $\mathrm{GF}(2^{64})$, evaluation at Q identifies $L(D)$ with $\mathrm{GF}(2^{64})$. At each rational point P, take the value of a function, or for $P \in \{P_\infty, P_a\}$ its leading Laurent coefficient. These 33 functionals are multiplicative and K-linear, so they give Eq. (2) with

$A_i = K$ for every i, provided they are independent on $L(2D)$.

The shift by P_a is needed for independence. With $D = 17P_\infty$, the function $x^{16} - x$ vanishes at all 32 affine points and lies in $L(32P_\infty)$, so it is killed by all 33 functionals. With the shifted divisor, the kernel is $L(2D - E)$, where E is the sum of all rational points. Since $2D - E \sim 2P_a - P_\infty$ has degree 1, the kernel is nonzero only if $2P_a \sim P_\infty + R$ for some point R. Then $\dim L(2P_a) \geq 2$, so $2P_a$ would be the canonical class. That happens only when P_a is fixed by the hyperelliptic involution $y \mapsto y + 1$, and no affine point is. Shifted and asymmetric divisors of this kind are standard in Chudnovsky-type algorithms [20, 21]. The construction gives $\mathcal{T}(64) \leq 33 \times 39 = 1287$, and reduction lowers it to 1255. Numerically, the 33×33 evaluation matrix is invertible for all 32 choices of P_a, and the product identity was checked on random inputs before the factors were emitted; the certified decomposition does not depend on these checks.

When curves help. A curve of genus g needs $2d + g - 1$ rational units instead of $2d - 1$. It beats the projective line only when the line runs out of points and the places that would replace them cost much more per unit than a rational one. Over GF(16), a degree-2 place costs 113 for 2 units, against $2 \times 39 = 78$. Over $\mathbb{F}_2$, GF(4) and GF(8) the local rings are cheap substitutes. A cost model with the blocks of Table II finds that no curve of positive genus beats the line for the other sizes in Table I. The same model predicts gains over GF(16) for $n = 44$, 48, 52, 56 and 60. These are estimates, not certified circuits.

V. REDUCTION AND ANNEALING

The decompositions of a fixed signature tensor form a coset of the code $\mathcal{C} = \{w \in \mathbb{F}_2^{2^N-1} : \sum_y w_y y^{\otimes 3} = 0\}$, the punctured Reed–Muller code $\mathrm{RM}(N-4, N)^*$ [3]. A composed circuit is one point of this coset, and its weight is the T-count. We lower the weight in two ways.

FastTODD-type reduction. Our C implementation of FastTODD [5] looks for a vector z and a set of factors such that adding z to each of them preserves the tensor and creates equal pairs, which cancel. It accepts neutral moves with a fixed probability and handles up to $N = 192$ variables. On the composed circuits it removes a few gates and then stalls on a plateau (Table I). For a circuit of b blocks it first removes exactly $b - 1$ gates at $n = 11$–16, 32 and 64, for example 32 for the 33 blocks at $n = 64$. Long plateau walks can remove more. The five-block circuits for $n = 6$ and 10 lose $b + 1 = 6$, the latter after about a minute. The three-block circuit for $n = 8$ loses 4, but only in about a third of short runs. At $n = 11$, one walk of 900 CPU seconds left the 181-gate plateau after 188 seconds and reached 179. Two walks from other $n = 11$ networks did not. The pattern $\sum_k \mathcal{T}(A_k) - (b-1)$ mirrors the exact count $7m - (m-1) = 6m + 1$ for m disjoint CCZ gates [23]. The smallest case, $\mathcal{T}(\mathbb{F}_2 \times \mathbb{F}_2) = 13 < 2 \times 7$, also follows from the exhaustive enumeration of Sec. VI. Blocks combined in one circuit can apparently share about one gate each. We state this only as an observation. Elsewhere the plateaus have held. At $n = 12$, 13, 16 and 64, 8 to 30 short runs over seeds and gauges (field isomorphisms and point choices) end at the same count. Walks of 15 to 30 minutes from the raw circuits for $n = 8$, 10, 12, 13, 14, 16 and 32 also end at the counts of Table I. So does a 20-minute walk at $n = 64$, but at that size it covers only 2×10^4 trials.

Annealing over flats. For every affine 4-flat $F \subset \mathbb{F}_2^N$, $\sum_{v \in F} v^{\otimes 3} = 0$. The singles, pairs and triples restricted to F are polynomials of degree ≤ 3, and their sum over a 4-flat vanishes. So toggling the nonzero points of F in the decomposition is a move within the coset. We run Metropolis sampling on the count with proposals spanned by members of the current decomposition, so that many points of F are already present and the move is not too costly. At $n = 7$, a fixed temperature of 0.30 (in units of one gate) for 8×10^8 steps took the start 111 to 100. Restarting from each new best with a new seed at a slightly higher temperature (0.34, then 0.32) gave 99 and then 97. At $n = 6$, the start 85 went to 79 by FastTODD and to 77 at fixed temperature. Further fixed-temperature runs never left 77. A linear cooling schedule from 0.40 to 0.27 over 1.2×10^9 steps reached 71 in one of three seeds (the others ended at 77 and 76). A single hot restart from 71 (0.65 to 0.28) reached 69 within seconds. Cooling broke a floor that longer fixed-temperature runs did not, but these are few runs.

Annealing did not lower the best counts for $N \geq 24$. Cooling runs from $n = 8$: 113, $n = 9$: 139, $n = 10$: 151 and $n = 11$: 181 (before the long walk), with several schedules and up to 1.5×10^9 steps per run, all ended where they started. So did a single hot run at $n = 12$: 191. At these sizes most proposals raise the count by 6 or more. A temperature low enough to keep the walk near the floor rejects almost all of them, and one high enough to move it destroys the structure.

VI. RESULTS

Records. Table I lists the thirteen certified circuits and how each was obtained. The reductions relative to the reported values grow with n: 4–10% for $n = 6$–10, 16–30% for $n = 11$–16, 51% for $n = 32$ and 66% for $n = 64$. For $n = 5$ the circuit is two gates below the checked value. The moduli are listed in Appendix A. All the starts are deterministic and take well under a second to build. The searches are seeded, and their outputs are stored.

Optimality for $n = 2$ and for $\mathbb{F}_2[y]/(y^2)$. For GF(4) ($N = 6$), the code $\mathcal{C}$ has dimension 22. Enumerating all 2^{22} coset elements gives minimum weight 17, attained by 1008 decompositions, with the next weights 18 (2576) and 21 (24 816). Hence $\mathcal{T}(2) = 17$ in the model of Sec. II, which the published circuits already attain [5, 6]. A second implementation, written indepen-

dently, reproduced these numbers. The same enumeration shows that $\mathcal{T}(\mathbb{F}_2[y]/(y^2)) = 15$ (112 minimizers) and $\mathcal{T}(\mathbb{F}_2 \times \mathbb{F}_2) = 13$. The auditor agent reproduced both with its own code. These are optima within the model, not lower bounds for unrestricted circuits.

Small fields. For $n = 3$, 4 and 9 our certified values equal the best published values with correct released data: $\mathcal{T}(3) \leq 29$ and $\mathcal{T}(4) \leq 39$, as in Ref. [6], and $\mathcal{T}(9) \leq 139$, as in Ref. [8] (Khoruzhii start). For $n = 5$ we certify $\mathcal{T}(5) \leq 57$. This beats the 59 of Refs. [6, 7] but not the reported 49 [8], whose released data is not a GF(32) multiplication tensor (Sec. VI). Two of the reported small values deserve comment.

For $n = 3$, Ref. [5] reports 23 and Ref. [8] 27. The 27 starts from a matrix that is not a GF(8) multiplication tensor (Sec. VI). We could not reproduce the 23. Vandaele's public synthesis code [13] produces a 12-qubit GF(8) circuit with 8 CCZ gates and three clean ancillas. His public optimizer [5] reduces it to 31 T gates; we simulated the result, and it is a correct in-place multiplier when the ancillas start at 0. The 6-CCZ start behind the published 23 is in neither repository. Our own searches are just as stuck. More than 3000 FastTODD-type and greedy descents, from every field start we tried, end at exactly 29. These include our rebuild of the Karatsuba construction behind the start of Ref. [8]. Our rebuild has the same 40 gates as theirs but, unlike theirs, it is a valid GF(8) tensor. A Stern information-set-decoding search of the full coset found valid decompositions at weights 33 down to 29 at the rates a heuristic random-code model predicts. It found none of weight ≤ 28, where that model predicts about four. Exact searches over decompositions invariant under several symmetry groups found nothing of weight ≤ 27. The public benchmark repository of Ref. [5] contains GF(2^n) circuits for $n \geq 4$ only, so the $n = 3$ circuit cannot be checked from source. The same Karatsuba start reduces to exactly 23 for some reducible moduli, for example $x^3 + 1$, where the algebra is $\mathbb{F}_2 \times$ GF(4) rather than a field. This suggests, but does not prove, that 23 is not a GF(8) value. Search stagnation, exhausted symmetric families and a heuristic model are not lower bounds, so none of this refutes the published values. We therefore keep 23 as the record in our verifier, and we do not use it as a base.

For $n = 5$, the reported 49 [8] comes from the same family of starts as the 27 and does not decompose GF(32) multiplication (Sec. VI). Our 57 comes from CRT over places of $\mathbb{F}_2$: ∞^3, x^2, $(x+1)^2$ and one degree-2 place, 72 gates. Annealing at fixed temperature reaches 60, and a reheat at temperature 0.30 reaches 58 after 4×10^8 steps. FastTODD-type reduction of the 58 then gives 57, in each of four seeds.

Bounds implied by published values. Theorem 1 also applies to published small values, but only if they are genuine. A 49-gate GF(32) block would give $\mathcal{T}(10) \leq 3 \times 49 = 147$, below our 151. We downloaded the released 49-gate matrix to build this circuit, and found that it is not a GF(32) block (Sec. VI). So it implies no bound. A 23-gate GF(8) block, if it exists, would give $\mathcal{T}(9) \leq 115$, $\mathcal{T}(12) \leq 161$ and $\mathcal{T}(15) \leq 207$. These bounds remain conditional. Our $n = 6$ record of 69 equals the bound 3×23 but does not depend on it.

Published values and their released data. A released decomposition fixes its signature tensor. It decomposes GF(2^n) multiplication, up to linear maps on the three registers, if and only if three conditions hold. First, its singles and pairs are even. Second, its odd triples have the form (a_i, b_j, c_k). Third, the bilinear map $M(a, b)$ they define is isotopic to field multiplication. A field has no zero divisors, and isotopy preserves that. So if some $a \neq 0$ makes the map $b \mapsto M(a, b)$ singular, M is not GF(2^n) multiplication in any basis.

We applied this test to the released data of Ref. [8]. Its final matrices and the starting matrices of its search code are public under the MIT license. Every start derived from Vandaele's circuits ($n = 3$–16 and 32) and every final matrix of that branch ($n = 5$, 6, 9–16 and 32) satisfies the first two conditions but has zero divisors (Table III). In these starts the products $x^0, \ldots, x^{n-1}$ map correctly, but some reductions of higher powers are wrong. For example, at $n = 5$ with $P = x^5 + x^2 + 1$ the start maps $x^6 \mapsto 1 + x + x^3$ instead of $x + x^3$. In 15 of the 21 released starts the error follows one rule. The reductions follow the correct recursion $x^{m+1} \mapsto x \cdot (x^m \bmod P)$, but a spurious constant term enters at $m = 2(n - k)$ for each middle term x^k of P. The other six use moduli with a middle exponent $k > n/2$, and their errors take a different form. For $n = 5$–16 each released final encodes the same map as the corresponding released start, for example the $n = 5$ start (98 gates) and final (49 gates). The $n = 32$ final matches none of the three released $n = 32$ starts, but it has zero divisors as well. The values of this branch in Table I of Ref. [8], among them $n = 5$: 49 and $n = 11$–16 and 32: 215–1177, are therefore not T-counts of GF(2^n) multipliers. The $n = 64$ final (3729) was not released, but its released start (5103 gates) has the same defect. The Khoruzhii-start finals ($n = 8$: 123, 9: 139, 10: 157) pass: we mapped them to our basis, and the verifier accepts them. The audit agent replicated every count in Table III with independently written code. The scripts, the data hashes and the full output are in the archive.

VII. VERIFICATION

Frozen verifier. A candidate lists n, an irreducible modulus and the vectors u_r. The verifier checks irreducibility with Rabin's test, and then the criterion of Lemma 1 exactly, through the parities of all singles, pairs and triples. For each pair (i, j) it accumulates the XOR of the vectors containing both, so the cost is quadratic in the vector weight. By Lemma 1, a pass is a proof of correctness. A candidate wins only if it has fewer vectors than the record. Before the search started, the verifier's sources, the record table and the compiled checker were

TABLE III. Released matrices of Ref. [8] from the Vandaele start (starting matrix and final matrix, with their gate counts): the number of nonzero a for which $b \mapsto M(a, b)$ is singular. A GF(2^n) multiplication tensor has none. For $n = 32$ and 64, 4000 random a were tested. The Khoruzhii-start matrices ($n = 3$–10, and the finals for $n = 8$, 9, 10) have none, and the three finals pass our verifier after mapping. The released $n = 16$ final has 385 gates; Table I of Ref. [8] gives 383.

n	start	singular	final	singular
3	40, 42	2/7	(27)	–
4	56	4/15	(39)	–
5	98	12/31	49	12/31
6	132	14/63	79	14/63
7	164	32/127	–	–
8	189	148/255	–	–
9	320	240/511	165	240/511
10	334	550/1023	187	550/1023
11	376	852/2047	215	852/2047
12	396	2064/4095	243	2064/4095
13	460	3978/8191	293	3978/8191
14	492	8406/16383	319	8406/16383
15	536	8192/32767	363	8192/32767
16	567	32762/65535	385	32762/65535
32	1701	2023/4000	1177	2050/4000
64	5103	2023/4000	(3729)	–

pinned by hash. They were re-pinned three times: after the record table was updated, after robustness fixes (Sec. VIII), and for the quadratic-cost tensor check. The verifier accepts all nine published AlphaTensor-Quantum decompositions for $n = 2$–10, accepts schoolbook CCZ constructions, rejects malformed input and exercises every verdict path. Its 37 tests include random property tests. All single-bit corruptions are rejected, and all zero-sum moves are accepted.

Independent circuit check. A separately written C program evaluates $4f_n(x) - \sum_r \ell_{u_r}(x) \bmod 8$ for all 2^{3n} inputs. It checks that the remainder is a diagonal Clifford phase, using a Walsh–Hadamard transform and a Möbius transform over $\mathbb{Z}_4$. The verifier runs it for $n \le 8$, and the two checks have never disagreed. The auditor agent (Sec. VIII) simulated every input for the records with $n = 6$, 7, 10, 11 and 12, the last over $2^{36} \approx 6.9 \times 10^{10}$ inputs, with no failure. So every record with $n \le 12$ is certified by full simulation as well.

Independent tensor checks. The coordinating session re-checked every new best with a third implementation. It uses numpy and shares no code with the verifier, and it builds $\sum_r u_r^{\otimes 3}$ and the target from scratch. Every record has zero mismatches among the N^3 entries, for instance 7 077 888 entries at $n = 64$. The auditor re-checked each record with its own tensor code and a randomized circuit test. Negative controls fail every check. These include single flipped bits and the seven nonzero vectors of a 3-dimensional subspace, which the weight-≤ 2 part of the check cannot see.

Released proof objects. For each record we release the vectors u_r, the diagonal Clifford correction and an explicit OpenQASM gate list (Data availability). The correction is determined by the vectors: by the proof of Lemma 1 it is $\prod_i S_i^{-N_i/2}$ times a CZ gate on each pair (i, j) with $N_{ij} \equiv 2 \pmod 4$, where N_i and N_{ij} count the vectors containing i and $\{i, j\}$. A standalone checker, using only the Python standard library and sharing no code with the verifier, reads each gate list. It tracks the linear form carried by each wire through the CNOT gates and adds the exact $\mathbb{Z}_8$ phase polynomial of every T, S, Z and CZ gate. It then requires the sum to equal $4f_n$ coefficient by coefficient, with every wire returned to its input. This proves the released circuit correct for all inputs at every n. The checker also simulates each circuit on all 2^{3n} basis states for $n \le 8$ and on 2048 random ones otherwise, and checks the blocks and the $n = 64$ curve data. The checker also verifies a hash manifest and requires exactly the records and blocks claimed here. Twenty-five negative controls are all rejected. They drop a CZ or a CNOT, turn a T into $\mathrm{T}^\dagger$, flip a vector bit, swap the modulus, and include the malformed files found by red-teaming the checker.

Novelty. We found no published T-count at or below ours for the twelve field sizes $n = 6$–8, 10–16, 32 and 64. For $n = 5$, the only lower reported value is the one discussed in Sec. VI. The interpolation, CRT and curve constructions are classical in algebraic complexity [14–18, 20, 21]. We claim their combination with T-optimized blocks and the resulting circuits, not the constructions themselves. CRT-based multipliers have been used to lower Toffoli counts for binary fields with $n = 233$–571 [22]. We found no prior use of curve-based multiplication for T or Toffoli counts.

VIII. HOW THE RESULTS WERE FOUND

The work used three runs of a multi-agent system of language-model agents (Claude Sonnet 5.5, Anthropic), run as non-interactive coding sessions on one laptop with ten cores. A supervising session (Claude Opus 5.5) coordinated them. The agents shared a message board and could not modify the verifier, the record table or the board software. Every submission was frozen and verified, and the verifier, not the submitting agent, posted the verdict.

First run (subfields). Five agents took the roles of TODD-type reduction, bilinear construction, local search, large fields and audit. Six minutes after launch, the auditor found Ref. [8], which had lowered several of the records we had started from. The table was updated and re-frozen. Two wins certified in the minutes between the alert and the update ($n = 7$: 103, $n = 10$: 193) were revoked as superseded. The auditor also found three robustness bugs in the verifier and gaps in the board's

identity model. None could produce a false win, and all were fixed. The large-field agent proposed Karatsuba over subfields 29 minutes after launch. The coordinating session suggested the interpolation generalization, and the first six composite records followed within 27 minutes.

Second run (places, curves, annealing). Six agents ran for four hours in 45-minute shifts. Their roles were reduction, three small-field agents ($n = 3$, 4 and 5, and their multiples), composition and audit. Two minutes after launch, the reduction agent lowered $n = 8$ to 113, and a minute later $n = 32$ to 571. A build for 192 variables reduced the first $n = 64$ circuit after 19 minutes. At seven minutes, the agent for $n = 4$ proposed CRT over places with local rings, giving the first prime-n records ($n = 11$: 189, $n = 13$: 228). At 18 minutes the dual-number places over GF(4) gave $n = 10$: 151. The composition agent wrote the dynamic program over places, and adding the p^2 place brought $n = 11$ and 13 to 181 and 219 after 29 minutes. The annealing chain for $n = 7$ ran after 39–44 minutes, and the genus-2 curve for $n = 64$ came after 50–53 minutes. Slow cooling for $n = 6$ came after 73 minutes (71) and a hot restart after 102 minutes (69). The agents also shared failures. The reheating chain did not transfer to $N \geq 24$, cooling stalled at $n = 8$–12, and fixed temperatures failed at $n = 6$. These reports shaped the directives of the coordinating session.

The auditor's role changed the paper as well as the record table. It checked the first version of this manuscript against the board and found stale claims and two errors. It also prompted the investigation of the $n = 3$ value, reading the benchmark repository and the original papers.

Third run (review and release). A review of the first public version of this paper asked for the decompositions behind the records. It also pointed out that the published 49-gate GF(32) value would imply $\mathcal{T}(10) \leq 147$. The coordinating session built the release described in Sec. VII, downloaded the released matrices of Ref. [8] to compose that bound, and found the defect of Sec. VI instead. It also ran Vandaele's public code. Five agents then ran for three hours. Four minutes in, the reduction agent applied FastTODD-type reduction to the annealed 58-gate GF(32) decomposition and reached 57. In the last hour it found that the reduction plateaus are not always floors. A single 900-second walk took $n = 11$ from 181 to 179, a record that the audit agent checked on all 2^{33} inputs. The audit agent replicated the published-value audit with its own code and derived the defect rule. It red-teamed the release checker over six rounds, finding malformed bundles that the checker accepted, and every case was fixed before release. It also regenerated the $n = 64$ start from the released data alone. With its own code it re-derived the exact phase polynomial of all thirteen released circuits. It also simulated each circuit with $n \leq 16$ on random inputs, and found the correct output with phase $+1$ every time. The other searches found nothing new. These included exact symmetric searches for a 38-gate GF(16) decomposition, information-set decoding for a 24-gate $\mathbb{F}_2[y]/(y^3)$ block, hot annealing and walk-then-reduce schemes at $n = 3$–16, and alternative GF(16) bases in the composites. All thirteen records have odd T-counts. On every even-count state we tried, the reduction removed exactly one gate.

IX. OUTLOOK

Four directions follow. First, every improvement of a block in Table II lowers all the circuits that use it. For example, $\mathcal{T}(\mathbb{F}_2[y]/(y^3)) \leq 24$ would lower $n = 7$, 11 and 13, and $\mathcal{T}(\mathrm{GF}(16)) \leq 38$ would lower all eight circuits that contain GF(16) blocks. Second, the curve method applies whenever the line runs out of points. A certified $n = 44$–60 series would test the cost model. Third, annealing broke the floors that reduction left at $n = 6$ and 7 but none for $N \geq 24$; better move sets are needed there. Fourth, lower bounds remain far away. For Hadamard-free phase circuits with clean ancillas, and hence for multipliers of our form $H_c D H_c$ but not for unrestricted circuits, Ref. [23] proves $\mathcal{T}(n) \geq 6n + 1$ and recovers $\mathcal{T}(2) = 17$. Our upper bounds are 17.8 T gates per bit at $n = 32$ and 19.6 at $n = 64$. Exhaustive enumeration is infeasible already at $n = 3$, where $\dim \mathcal{C} = 382$. The status of $\mathcal{T}(3)$ (23, 27 or 29) is open.

ACKNOWLEDGMENTS

We thank the authors of Refs. [5–8] for publishing benchmark values and, in the case of Ref. [6], their decompositions (CC-BY 4.0), which serve as base cases. The circuits were found by an autonomous multi-agent system built on Claude Sonnet 5.5 (Anthropic). The verification software, the analysis, and this manuscript were produced by Claude Opus 5.5 (Anthropic).

DATA AVAILABILITY

The proof objects are archived with this paper at https://github.com/maxxfan/aixmathxphysics/tree/tcount-gf2-v3/papers/tcount-gf2-multipliers/artifacts (git tag `tcount-gf2-v3`). The archive contains, for each of the thirteen records, the modulus, the vectors u_r, the Clifford correction, the full OpenQASM 2.0 circuit and its gate counts. It also contains the block decompositions of Table II, and the $n = 64$ construction data: the curve, the point P_a, the place Q, the 33 evaluation functionals and the coefficients ω_k, with the identity they satisfy. Finally it contains the standalone checker with its expected output, the negative controls, the frozen verifier and a SHA-256 manifest. Running `python3 -I check_release.py .` re-checks, in about ten seconds, every T-count in the "This work" column

of Table I together with the circuits behind it. It does not certify the published comparison values. The composed starts are deterministic. The $n = 64$ start is regenerated bit for bit from the released construction data by `construction/regen_n64.py`. The reductions and annealing runs are seeded but time-limited, so the released vectors, not the search logs, are the record of the result.

TABLE IV. Moduli of the certified circuits.

n	P_n	n	P_n
5	$x^5 + x^2 + 1$	12	$x^{12} + x^3 + 1$
6	$x^6 + x + 1$	13	$x^{13} + x^5 + x^2 + x + 1$
7	$x^7 + x + 1$	14	$x^{14} + x^5 + 1$
8	$x^8 + x^4 + x^3 + x + 1$	15	$x^{15} + x + 1$
10	$x^{10} + x^3 + 1$	16	$x^{16} + x^5 + x^3 + x + 1$
11	$x^{11} + x^2 + 1$	32	$x^{32} + x^7 + x^3 + x^2 + 1$
		64	$x^{64} + x^{11} + x^2 + x + 1$

Appendix A: Construction data

Each record is specified by its modulus P_n and the construction in Table I. The decompositions, lists of vectors in $\mathbb{F}_2^{3n}$, and the circuits are in the archive (Data availability). Table V gives the gate counts of the released circuits. They are ancilla-free on $3n$ qubits. Each phase gadget uses a naive CNOT ladder, so the CNOT counts are upper bounds that we have not optimized.

TABLE V. Gate counts of the released circuits: qubits, T gates, CNOT gates (naive ladders), single-qubit diagonal Cliffords (S, $S^\dagger$, Z), CZ gates and Hadamards.

n	qubits	T	CNOT	$S, S^\dagger, Z$	CZ	H
5	15	57	746	13	49	10
6	18	69	1082	15	73	12
7	21	97	1854	13	112	14
8	24	113	2202	18	114	16
10	30	151	3938	25	203	20
11	33	179	5394	24	254	22
12	36	191	6006	26	236	24
13	39	219	7962	34	351	26
14	42	229	8482	34	420	28
15	45	253	10218	37	485	30
16	48	267	11754	33	555	32
32	96	571	51814	75	2310	64
64	192	1255	238466	151	9190	128

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